

Psychology of Loneliness

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ABSTRACT

Loneliness is a subjective distress arising from a perceived mismatch between desired and actual social relationships and has been consistently linked to adverse mental and physical health outcomes. Despite the growing recognition of its public health impact, empirical investigations among Indian populations remain scarce. This study examined the prevalence, psychosocial correlates, and predictors of loneliness in a cross-sectional sample of university staff and students in Vadodara, India. Specifically, we aimed to (1) quantify loneliness levels using the UCLA Loneliness Scale (Version 3), (2) evaluate associations between loneliness, depressive symptoms (Beck Depression Inventory-II), perceived social support (MSPSS of Perceived Social Support), and maladaptive social cognitions (ULCS-SF), and (3) identify demographic and psychosocial predictors of loneliness via hierarchical regression. Between January and March 2025, 450 participants (age range 18–65 years; $M = 29.4$, $SD = 10.2$; 58% female) completed an online survey after providing informed consent following Sigma University IRB approval. The measures included the 20-item UCLA Loneliness Scale ($\alpha = 0.92$), the 21-item Beck Depression Inventory-II ($\alpha = 0.89$), the 12-item Multidimensional Scale of Perceived Social Support ($\alpha = 0.93$), and the 15-item UCLA Loneliness Cognitions Scale-Short Form ($\alpha = 0.88$), alongside demographic items on living arrangements and social media use. Data were analysed using SPSS v.26. Descriptive statistics characterised the sample, Pearson correlations assessed bivariate relationships, and hierarchical multiple regression determined the incremental predictive value of depressive symptoms, social support, and maladaptive cognitions on loneliness, controlling for demographics. Participants reported a mean loneliness score of 43.6 ($SD = 12.5$; range 20–78), with 32% scoring above the scale midpoint, indicating moderate-to-high loneliness. The mean BDI-II scores indicated mild depressive symptoms ($M = 14.2$, $SD = 9.1$), MSPSS scores reflected

moderate perceived support ($M = 58.3$, $SD = 14.8$), and ULCS-SF scores suggested moderate endorsement of negative social cognitions ($M = 38.7$, $SD = 10.3$). Correlational analyses revealed that loneliness was strongly positively associated with depressive symptoms ($r = 0.62$, $p < 0.001$) and maladaptive cognitions ($r = 0.71$, $p < 0.001$), and negatively associated with perceived social support ($r = -0.58$, $p < 0.001$). Individuals living alone reported significantly higher loneliness than those cohabiting ($t(448) = 5.12$, $P < 0.001$). Hierarchical regression demonstrated that demographic variables (age, gender, student/staff status, living arrangement) explained 8% of the variance in loneliness ($F(4,445) = 9.67$, $p < 0.001$), with living alone as a significant predictor ($\beta = 0.28$, $p < 0.001$). Adding depressive symptoms increased the explained variance by 31% ($\beta = 0.47$, $p < 0.001$), perceived social support added 20% ($\beta = -0.40$, $p < 0.001$), and maladaptive cognitions contributed an additional 10% ($\beta = 0.35$, $p < 0.001$), culminating in a total R^2 of 0.69 ($F(7,442) = 147.6$, $p < 0.001$). These findings underscore the multifactorial nature of loneliness, highlighting the cognitive, affective, and social support dimensions as key intervention targets. The prominence of maladaptive social cognitions supports cognitive-behavioural frameworks for loneliness reduction, while the protective role of perceived support advocates for structured peer-support programs, particularly among individuals living alone. Given the cross-sectional design and single-institution sample, causal inferences are limited, and generalisability is constrained. Future research should employ longitudinal designs to elucidate temporal dynamics, test the efficacy of tailored cognitive-behavioural and digital Ecological Momentary Interventions, and explore cultural moderators of loneliness in diverse Indian contexts. In conclusion, addressing loneliness in university settings requires integrative approaches that combine cognitive restructuring, social support enhancement, and technology-mediated real-time interventions. Such efforts may mitigate the substantial psychological burden of loneliness and promote well-being among young and middle-aged Indian adults.

Keywords: Loneliness; Depression; Perceived Social Support; Maladaptive Social Cognitions; Living Alone; University Population; Cognitive-Behavioral Therapy; Ecological Momentary Assessment

1. Introduction

Loneliness is a pervasive subjective experience characterised by discrepancies between desired and actual social relationships [1]. Unlike objective social

isolation, which refers to the quantifiable state of having few social connections, loneliness reflects an individual's appraisal of relationship quality and fulfilment [2]. Over the past decades, a growing body

of research has identified loneliness as a critical predictor of mental health disorders (e.g. depression and anxiety), physical morbidity (e.g. cardiovascular disease), and even early mortality [3,4].

The COVID-19 pandemic, with its attendant social distancing measures, has further highlighted loneliness as a global public health concern, affecting individuals across their lifespan [5]. Adolescents have reported heightened feelings of disconnection in the context of remote schooling [6], while older adults have experienced exacerbated isolation due to restrictions on communal activities [7].

Despite its ubiquity and profound impact, the psychological mechanisms underlying loneliness and effective strategies for assessment and intervention remain incompletely understood. This study presents a cross-sectional analysis examining the prevalence, correlates, and psychological consequences of loneliness among Indian adults. We employed validated psychometric instruments, explored associations with affective and social-cognitive variables, and discussed implications for intervention. The specific objectives were as follows:

1. To assess the prevalence and severity of loneliness in a sample of Indian university staff and students.
2. To examine the relationships between loneliness, depressive symptoms, perceived social support, and maladaptive social cognition.
3. To identify demographic and psychosocial predictors of loneliness using regression analysis.
4. The implications for psychological intervention are discussed, and directions for future research are proposed.

2. Materials and Methods

2.1. Participants

A total of 450 participants (age range 18–65 years; $M = 29.4$, $SD = 10.2$; 58% female, 42% male) were recruited via stratified convenience sampling from

Sigma University in Vadodara. The inclusion criteria were as follows: (a) age ≥ 18 years, (b) fluency in English or Hindi, and (c) willingness to provide informed consent. The exclusion criteria were self-reported diagnosis of major cognitive impairment or active psychosis.

2.2. Measures

- UCLA Loneliness Scale (Version 3): A 20-item self-report measure assessing subjective feelings of loneliness and social isolation. Responses range from 1 (“Never”) to 4 (“Often”); higher scores indicate greater loneliness. The Cronbach’s α in the present sample was 0.92 [8].
- Beck Depression Inventory–II (BDI-II): A 21-item inventory measuring depressive symptom severity over the past two weeks. Items are scored 0–3; higher total scores indicate more severe depression symptoms. $\alpha = 0.89$ [9].
- Multidimensional Scale of Perceived Social Support (MSPSS): A 12-item scale assessing perceived support from family, friends, and significant others. Responses range from 1 (“Very strongly disagree”) to 7 (“Very strongly agree”); higher scores reflect greater perceived support. $\alpha = 0.93$ [10].
- UCLA Loneliness Cognitions Scale–Short Form (ULCS-SF): A 15-item measure capturing maladaptive social cognitions (e.g. hypervigilance to social threat and negative self-appraisals). Responses were measured on a 5-point Likert scale; $\alpha = 0.88$ [11].
- Demographic Questionnaire: Items on age, gender, academic role (student vs. staff), living arrangement (alone vs. with family/peers), and frequency of social media use.

2.3. Procedure

Data collection will be conducted between January and March 2025. The study protocol was approved by the Sigma University Institutional Review Board. Participants completed an online survey (hosted on Qualtrics) comprising these measures. Informed consent was obtained electronically before the survey

was initiated. The average completion time was 25 minutes, and the participants received no monetary compensation.

2.4. Data Analysis

Data were analysed using SPSS v.26. Descriptive statistics were used to characterise the sample demographics and scale scores. Pearson's correlation coefficients were used to examine the associations among loneliness, depression, perceived social support, and maladaptive cognitions. Hierarchical multiple regression identified predictors of loneliness: Step 1 entered demographic variables; Step 2 added

BDI-II scores; Step 3 added the MSPSS; and Step 4 added the ULCS-SF. All tests were two-tailed ($\alpha = 0.05$).

3. Results

3.1. Participant Characteristics

Of the 450 respondents, 260 (58%) were female and 190 (42%) were male. Students comprised 62% ($n = 279$) and staff 38% ($n = 171$) of the participants. Living arrangements: 40% ($n = 180$) lived alone, and 60% ($n = 270$) lived with family or peers. The mean daily social media use was 3.2 h ($SD = 1.4$).

3.2. Descriptive Statistics for Key Variables

Table 1. Descriptive Statistics for Key Variables.

Measure	Mean (SD)	Range
UCLA Loneliness Scale (20-80)	43.6 (12.5)	20-78
BD-II (0-63)	14.2 (9.1)	0-52
MSPSS (12-84)	58.3 (14.8)	12-84
ULCS-SF (15-75)	38.7 (10.3)	15-73

3.3. Correlational Analyses

Pearson correlations (Table 2) revealed the following:

- Loneliness was positively correlated with depressive symptoms ($r = 0.62$, $p < 0.001$) and maladaptive social cognition ($r = 0.71$, $p < 0.001$).
- Loneliness was negatively correlated with perceived social support ($r = -0.58$, $p < 0.001$).
- Living alone was associated with higher loneliness ($t(448) = 5.12$, $p < 0.001$).

Table 2. Correlations among the key variables ($N = 450$).

Variable	1	2	3	4
1. Loneliness	--			
2. Depression	0.62***	--		
3. Perceived Social Support	-0.58***	-0.65***	--	
4. Maladaptive Social Cognitions (ULCS)	0.71***	0.59***	-0.54***	--

*** $p < 0.001$.

3.4. Regression Analyses

The hierarchical regression predicting loneliness (Table 3) showed:

- Step 1 (Demographics): Age, gender, student/staff status, and living arrangement explained 8% of the variance ($F(4,445) = 9.67$, $p < 0.001$), with living alone as a significant predictor ($\beta = 0.28$, $p < 0.001$).
- Step 2 (+ Depression): $\Delta R^2 = 0.31$ ($p < 0.001$); BDI-II was significant ($\beta = 0.47$, $p < 0.001$).
- Step 3 (+ Social Support): $\Delta R^2 = 0.20$ ($p < 0.001$); MSPSS significant ($\beta = -0.40$, $p < 0.001$).

- Step 4 (+ Maladaptive Cognitions): $\Delta R^2 = 0.10$ ($p < 0.001$); ULCS-SF significant ($\beta = 0.35$, $p < 0.001$). The overall model explained 69% of the variance in loneliness ($F(7,442) = 147.6$, $p < 0.001$).

Table 3. Hierarchical regression predicting loneliness.

Predictor	β (Step 1)	β (Step 2)	β (Step 3)	β (Step 4)
Age	-0.10*	-0.08	-0.05	-0.04
Gender (female = 1)	0.06	0.04	0.03	0.02
Student status (student = 1)	0.12*	0.08	0.06	0.04
Living alone (yes = 1)	0.28***	0.21***	0.18***	0.15***
BDI-II	--	0.47***	0.29***	0.21***
MSPSS	--	--	-0.40***	-0.25***
ULCS-SF	--	--	--	0.35***

$p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

4. Discussion

4.1. Interpretation of Key Findings

The present study found that loneliness levels among Indian university staff and students ($M = 43.6$) were comparable to those in Western samples [8,12]. Living alone emerged as a consistent demographic risk factor, aligning with prior studies linking solitary living to higher loneliness [13]. Psychosocial variables, particularly depressive symptoms, low perceived social support, and maladaptive social cognitions, accounted for most of the variance, underscoring the multifactorial nature of loneliness.

The strong correlation between loneliness and maladaptive cognitions ($r = 0.71$) supports the cognitive discrepancy model [14], suggesting that negative social self-schemas and hypervigilance to rejection fuel persistent loneliness. Moreover, perceived social support exerted a robust inverse association ($r = -0.58$), highlighting the protective role of supportive relationships [15].

4.2. Comparison with Previous Research

Our findings replicate the meta-analytic evidence that depression and loneliness are bidirectionally linked [3,16]. The incremental contribution of maladaptive cognition in predicting loneliness beyond depression and social support echoes Cacioppo and Hawkley's

evolutionary model, which emphasises cognitive calibration in response to perceived social threats [17].

Notably, age had a modest negative association ($\beta = -0.10$), indicating that younger adults may experience slightly higher loneliness in this sample than older adults. This contrasts with the typical U-shaped lifespan pattern, in which both adolescents and older adults report peaks [18]. The restricted age range and university context may explain this deviation.

4.3. Implications for Intervention

The predominance of cognitive and affective predictors suggests that cognitive-behavioral interventions targeting maladaptive social beliefs could yield significant reductions in loneliness [19]. Techniques such as cognitive restructuring, behavioural activation, and social skills training should be tailored to address hyper-vigilance and negative self-appraisals.

Enhancing perceived social support through group-based activities (e.g. peer mentorship and structured social groups) may buffer loneliness, particularly in individuals living alone. University counselling centres can implement peer-led support programs to foster connectedness.

Digital interventions, such as smartphone-based Ecological Momentary Interventions, could deliver in-the-moment cognitive reframing prompts, leveraging the real-time assessment capabilities demonstrated by EMA research [20].

5. Conclusion and Suggestions

This study contributes to the understanding of the psychological underpinnings of loneliness in an Indian university context. Key conclusions are:

1. High prevalence: Nearly one-third of the participants reported moderate to high loneliness.
2. Multifactorial predictors: Depression, low social support, and maladaptive cognitions jointly explained 69% of the variance in loneliness.
3. Living alone is a salient demographic risk factor.

The following suggestions for practice are provided:

- Integrative CBT programs focusing on social-cognitive biases.
- Peer support networks within university settings, especially for students living alone.
- Digital monitoring and intervention tools to provide timely support are also available.

6. Limitations and Directions for Future Research

6.1. Limitations

- The cross-sectional design precludes causal inference.
- Convenience sampling from a single university limits the generalisability of the findings to the broader Indian population.
- Self-report measures may be subject to social desirability and recall biases.

6.2. Future Research

1. Longitudinal studies disentangling the causal pathways between loneliness and mental health outcomes.
2. Experimental trials assessing the efficacy of tailored CBT and digital interventions in reducing loneliness.

3. Cross-cultural comparisons exploring how collectivistic norms in India modulate loneliness trajectories.
4. Qualitative investigations to capture lived experiences of loneliness across diverse socioeconomic groups.

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